Aether

Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

DATASET NAME	**Aether_Copilot_Clean_Hospital_100_People.csv**
REPORT DATE	10 October 2026, 05:49 UTC
DATASET VERSION	Version 1
TOTAL RECORDS	100 rows
FEATURE DIMENSIONS	17 columns (3 numeric, 13 categorical)
DATA QUALITY SCORE	**100/100**
SCOPE OF ANALYSIS	Full-Spectrum Exploratory & Statistical Audit

Notice: This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

Executive Summary

TOTAL RECORDS	TOTAL DIMENSIONS	DATA QUALITY SCORE
100	**17**	**100/100**
MISSING DATA RATE	DUPLICATE RECORDS	DETECTED OUTLIERS
0.0%	**0**	**0**

This report presents a structured analysis for **Aether_Copilot_Clean_Hospital_100_People.csv** (Version 1). The dataset comprises **100 observations** across **17 attributes**. Empirical evaluation yields an overall Data Quality Score of **100/100**, measuring record completeness (100%), uniqueness (100%), and validity (100%).

Key Summary Findings:

• The dataset contains 100 verified observations across 17 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'Age' and 'Billing_INR' (r = 0.08, little or no linear relationship).

• Categorical concentration observed in 'Gender' where 'Male' accounts for 53.0% of total entries.

Dataset Overview

DATASET NAME	**Aether_Copilot_Clean_Hospital_100_People.csv**	ANALYSIS DATE	10 October 2026, 05:49 UTC
ROW COUNT	100	COLUMN COUNT	17
FILE FORMAT	CSV	MEMORY FOOTPRINT	81.1 KB

Column Overview:

Column	Data Type	Non-null	Missing	Unique	Summary
Patient_ID	str	100	0 (0.0%)	100	100 unique identifiers
Patient_Name	str	100	0 (0.0%)	88	Top: Kavya Patil (3%)
Age	int64	100	0 (0.0%)	49	[20.0 to 82.0], Med: 50.0
Gender	str	100	0 (0.0%)	2	Top: Male (53%)
Blood_Group	str	100	0 (0.0%)	8	Top: A- (18%)
Department	str	100	0 (0.0%)	10	Top: Oncology (19%)
Primary_Condition	str	100	0 (0.0%)	10	Top: Kidney Disea (19%)
Admission_Date	str	100	0 (0.0%)	30	Top: 18/08/2026 (9%)
Discharge_Date	str	100	0 (0.0%)	34	Top: 27/08/2026 (8%)
Length_of_Stay_Days	int64	100	0 (0.0%)	12	[1.0 to 12.0], Med: 7.0
Admission_Type	str	100	0 (0.0%)	4	Top: OPD (27%)
Doctor	str	100	0 (0.0%)	6	Top: Dr. Arjun Me (20%)
Ward	str	100	0 (0.0%)	4	Top: Ward D (32%)
Billing_INR	int64	100	0 (0.0%)	100	[10800.0 to 94599.0], Med: 52604.0
Payment_Type	str	100	0 (0.0%)	3	Top: Self Pay (35%)
Treatment_Status	str	100	0 (0.0%)	4	Top: Discharged (32%)
Risk_Level	str	100	0 (0.0%)	3	Top: Medium (44%)

Data Quality

Quality Dimension	Score	Benchmark	Assessment Detail
Completeness	**100%**	>= 95.0%	Calculates proportion of non-null cells across the observation matrix.
Uniqueness	**100%**	>= 98.0%	Measures absence of identical duplicate row vectors.
Validity	**100%**	>= 90.0%	Evaluates presence of constant zero-variance features.
Consistency	**100%**	>= 85.0%	Checks for type homogeneity and uniform data formatting.

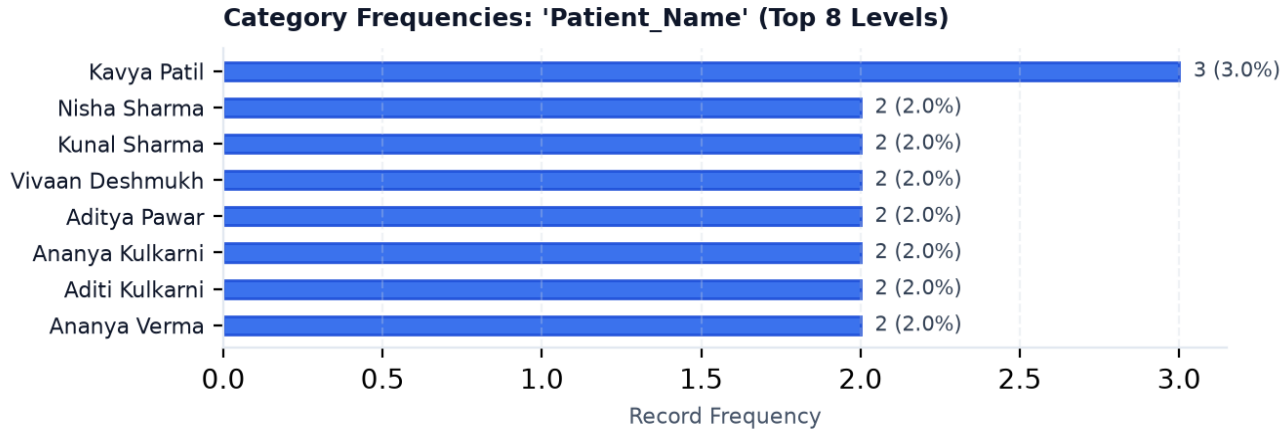
Audit Details: 0 missing cells identified (0.0% of dataset). Duplicate row analysis identified 0 duplicate record(s) (0.0%).

Descriptive Statistics

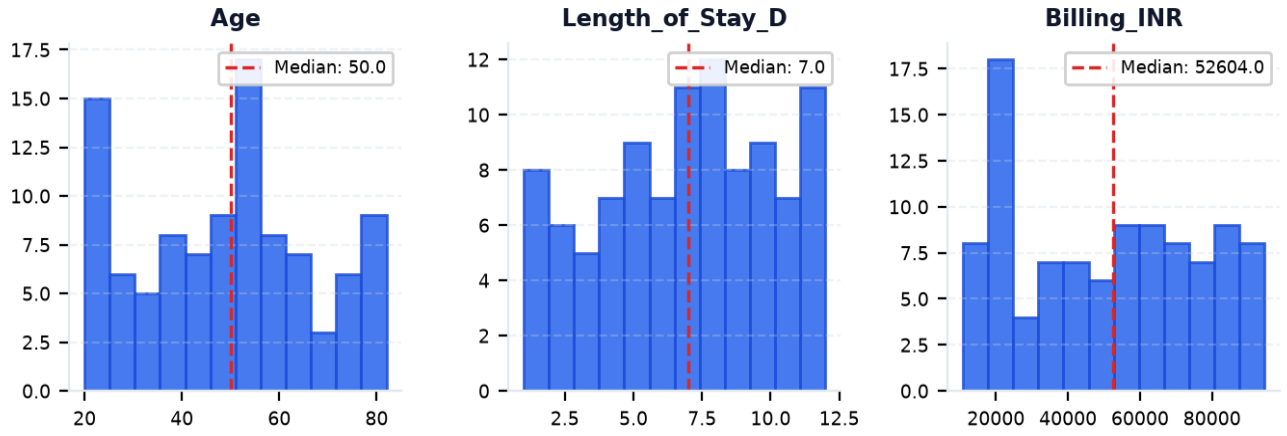
Feature	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Skew
Age	100	49.1	50.0	18.1	20.0	82.0	34.0	61.5	27.5	0.06
Length_of_Stay	100	6.9	7.0	3.4	1.0	12.0	4.0	10.0	6.0	-0.19
Billing_INR	100	51107.8	52604.0	25686.3	10800.0	94599.0	24012.2	70947.5	46935.2	0.06

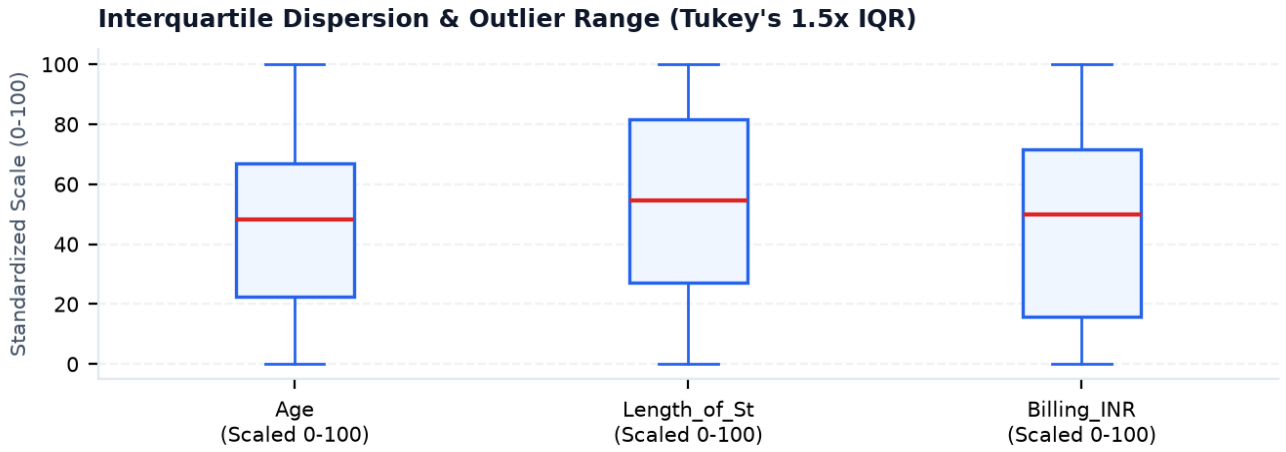
Categorical Analysis

Column	Unique	Most Frequent Value	Frequency	Percentage	Status
Patient_Name	88	Kavya Patil	3	3.0%	Balanced
Gender	2	Male	53	53.0%	Dominant (53.0%)
Blood_Group	8	A-	18	18.0%	Balanced
Department	10	Oncology	19	19.0%	Balanced
Primary_Condition	10	Kidney Disease	19	19.0%	Balanced
Admission_Date	30	18/08/2026	9	9.0%	Balanced
Discharge_Date	34	27/08/2026	8	8.0%	Balanced
Admission_Type	4	OPD	27	27.0%	Balanced

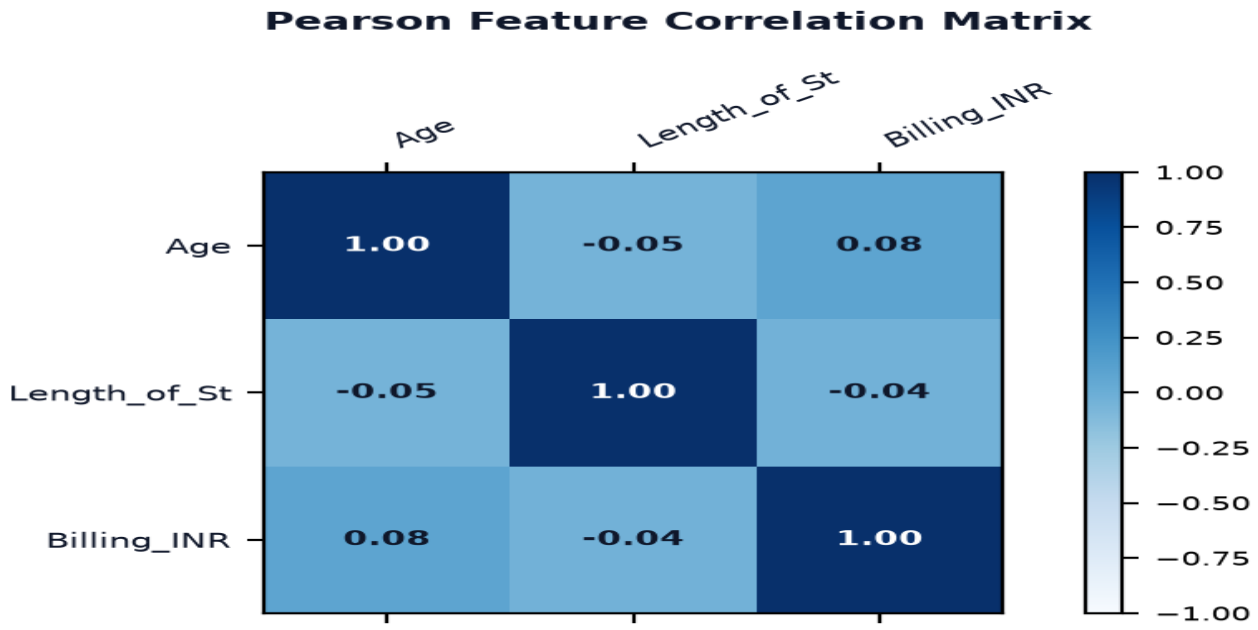


Distribution Analysis



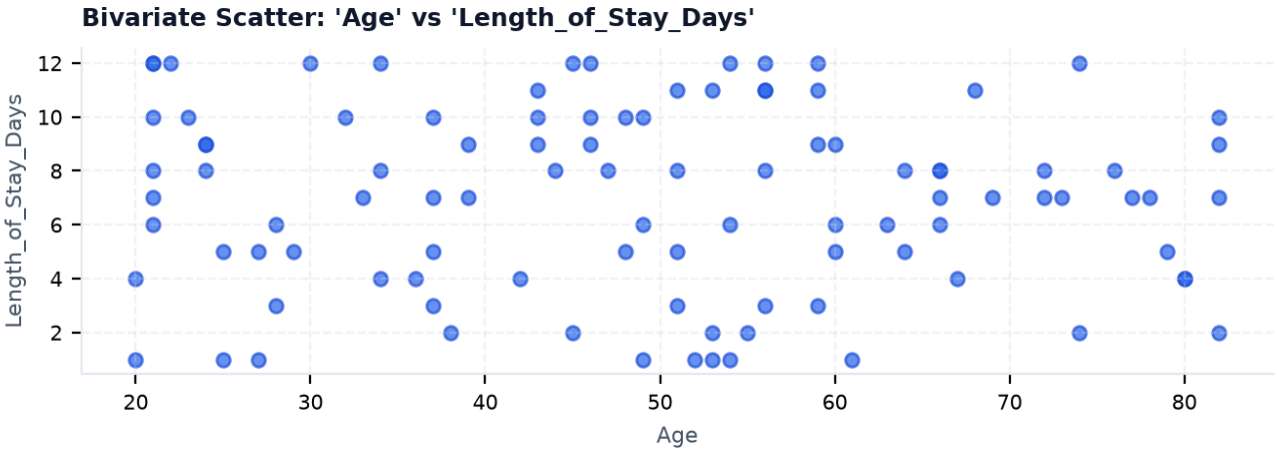

Interquartile Dispersion & Outlier Range (Tukey's 1.5x IQR)
Standardized Scale (0-100)
100
80
60
40
20
0
Age
(Scaled 0-100)
Length_of_St
(Scaled 0-100)
Billing_INR
(Scaled 0-100)

Correlation Analysis



Feature X	Feature Y	Pearson r	Strength	Direction
Age	**Billing_INR**	0.08	Little or no linear relationship	Positive
Age	**Length_of_Stay_Days**	-0.05	Little or no linear relationship	Positive
Length_of_Stay_Days	**Billing_INR**	-0.04	Little or no linear relationship	Positive

Relationships



Outlier Analysis

Feature	Outlier Count	Percentage	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)
Age	0	0.0%	-7.25	102.75
Length_of_Stay_Days	0	0.0%	-5.00	19.00
Billing_INR	0	0.0%	-46390.62	141350.38

No statistical outliers detected outside standard 1.5x IQR thresholds.

Key Findings

• The dataset contains 100 verified observations across 17 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'Age' and 'Billing_INR' (r = 0.08, little or no linear relationship).

• Categorical concentration observed in 'Gender' where 'Male' accounts for 53.0% of total entries.

Recommendations

Category	Priority	Action Recommendation
Operational	**Low**	Monitor the dominant category 'Male' in 'Gender' to identify potential sampling bias or operational dependency. Further domain validation is recommended.
Data Quality	**Low**	Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

Appendix

Mathematical Methodology	• **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR]. • **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)). • **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry.
Data Quality Scoring	Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows.
Compilation Provenance	Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.6 / NumPy 2.5.3. Timestamp: 10 October 2026, 05:49 UTC.